Epistemic Hygiene Arena-Uncued: A Diagnostic Pilot for LLM Agents in Polluted Evidence Environments

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Abstract

LLM systems increasingly reason over evidence environments that contain stale records, repost chains, generated pages, contradictory fragments, and metadata shortcuts. We introduce Epistemic Hygiene Arena-Uncued, a role-uncued diagnostic pilot for measuring evidence hygiene in these polluted evidence ecologies: final verdict correctness, provenance discipline, source independence, pollutant rejection, and executable verification behavior are scored separately. A previous cued construction exposed a benchmark-design failure: visible labels and document roles could leak the intended answer. That run is quarantined and is not used as evidence here. The new pilot removes direct material-role and conclusion cues from model-visible documents, checks leakage and shortcut baselines before model calls, and packages prompts, outputs, scorer artifacts, and structured-output diagnostics. In a 60-task, four-model pilot with two neutral metadata views, all model calls parse successfully, but answer correctness alone hides evidence-contract failures. Generated-lore rows illustrate the diagnostic value and the scoring risk: strict operational escape is low, yet an audit found that 71 of 72 target generated-lore strict failures close under prose-aware support-allocation sensitivity, leaving one likely true dirty-support failure. Shortcut baselines are competitive overall, bounding ecological and ranking claims. The release is a synthetic, pilot-scale diagnostic artifact, not deployment-ready evidence, an open-web benchmark, or a general leaderboard.

1 Introduction

Information-seeking systems are often evaluated by final-answer accuracy. That is too coarse for settings where the evidence environment is polluted. A system can answer correctly while citing a repost, laundering generated text into a support field, treating echo-chain records as independent, or failing to specify the next verification action. These are not only presentation errors: downstream tools, auditors, and users may depend on structured evidence and action fields.

Epistemic Hygiene Arena-Uncued tests evidence-hygiene behavior in a small controlled setting. The pilot asks models to judge claims using synthetic evidence packets that contain clean evidence, conflicting evidence, false consensus, buried primary evidence, or generated lore. The model-visible packet does not label documents as primary, contaminant, generated, or gold-supporting. The scorer then separates final verdict correctness from provenance discipline, source independence, pollutant rejection, and active-verification behavior.

This paper replaces an earlier cued construction. That older run was useful as a failure discovery: it showed that visible construction cues could invalidate the benchmark. It is quarantined and not used as a main result. The contribution here is a role-uncued synthetic diagnostic pilot with explicit pre-model leakage gates, shortcut baselines, four-model execution, scorer audit, narrow follow-up audits, and a reproducible artifact package.

The study is intentionally modest. It is a diagnostic pilot over 60 latent tasks, two neutral metadata views, four model labels, and one main prompt/schema condition. It should be read as evidence about scorer-contract behavior under controlled synthetic pollution, not as a stable ranking of providers, production readiness evidence, or an open-web ecological-validity claim.

The main contributions are:

- a role-uncued version of the Epistemic Hygiene Arena task design that removes direct material-role and conclusion cues from model-visible evidence;
- pre-model leakage checks and shortcut-baseline gates that bound which result claims are valid;
- a failure-decomposition and support-ambiguity analysis showing how verdict correctness, support fields, pollutant rejection, and verification actions can diverge under the current scorer contract;
- a packaged artifact, verifier, and scorer audit that make the pilot reproducible and explicitly bounded.

2 Polluted Evidence Environments

Retrieval and tool-use benchmarks have shown that language models can use external evidence and actions, but the quality of the evidence environment itself is often treated as a background assumption [1, 4, 5]. In open-web settings, that assumption is fragile. Evidence may be stale, copied from another source, contradicted by a primary record, or generated in a form that looks plausible. Fact-verification work asks whether claims are supported by evidence [3]; RAG evaluation and citation-faithfulness work ask whether generated answers remain grounded in retrieved material [2]; retrieval-poisoning work shows that retrieval itself can be adversarially corrupted [6]. Agent benchmarks add action and tool-use pressure [4, 5]. Epistemic Hygiene Arena-Uncued combines these concerns by scoring answer correctness, support hygiene, and executable verification actions inside a controlled polluted-evidence packet.

Epistemic Hygiene Arena focuses on polluted evidence environments rather than only polluted final answers. The target behavior is epistemic escape: the model should form the right belief, select safe support, reject or avoid polluted evidence, and, when needed, express an executable verification action.

The role-uncued pilot is built around five evidence conditions:

- **Clean**: the packet contains enough clean evidence and no intended pollutant trap.
- **Conflicting evidence**: the packet contains evidence that pulls toward incompatible conclusions.
- **False consensus**: several surface-consistent records point the wrong way.
- **Buried primary**: the primary evidence is present but less salient than distractors.
- **Generated lore**: generated or lore-like material appears plausible but should not be used as clean support.

These conditions are synthetic by design. The point is not to mimic the full web, but to make failure modes measurable under controlled source relations.

3 EHA-Uncued Dataset Design

The pilot contains 60 latent tasks. Each task is rendered in two neutral metadata views, giving 120 model-visible task instances per model. The condition split is balanced: 12 clean, 12 conflicting-evidence, 12 false-consensus, 12 buried-primary, and 12 generated-lore tasks. The family split is 24 packet-judgment, 24 evidence-selection, and 12 active-verification tasks.

The key design change is role uncuedness. Model-visible documents use opaque per-task identifiers such as `doc_001`; visible titles, source types, timestamps, bodies, and citations do not expose construction roles such as primary support, contaminant, generated document, stale document, or gold verdict. Gold labels, dependency edges, and audit-only role labels are preserved for scoring but are not placed in model-visible prompts.

The scorer distinguishes five paper-facing evidence roles. **Clean support** is evidence that can support the gold verdict. **Refuting evidence** supports the opposite verdict or a contradiction. **Polluted material** includes generated, stale, derivative, contaminant, or same-root echo evidence that should not be treated as independent support. **Rejected or diagnostic evidence** may be selected or discussed to explain why the environment is unsafe, but should not be allocated as clean support. **Verification action** is an executable next step, such as tracing a source relation or opening a primary record.

Two views test whether neutral metadata changes behavior without making roles explicit:

- `neutral_metadata_visible`: neutral metadata fields are visible.
- `neutral_metadata_hidden`: the same task is presented without those visible neutral metadata fields.

The pilot is not a replacement for a large benchmark. It is a validity-first dataset slice used to test whether the benchmark can survive basic leakage and shortcut checks before making model claims.

Two metadata qualifications matter for interpretation. First, document timestamps are synthetic, model-visible, and non-evidential; they should not be read as publication dates, observation dates, recency cues, or answer evidence. Second, false-consensus episodes may include mutual visible-citation loops among polluted same-root echo documents. These loops are synthetic echo-chain structure, not independent corroboration and not acyclic provenance trees.

Appendix C gives concrete task, packet, output, and scorer examples, including a generated-lore row where the model reaches the correct final verdict but fails strict operational escape by retaining polluted support. Appendix E reports the synthetic-template diversity audit and its limitations.

4 Leakage Controls And Shortcut Baselines

The older cued construction is quarantined because it exposed the wrong lesson: if benchmark construction roles are visible, models may solve the interface rather than the epistemic problem. The role-uncued pilot therefore runs leakage checks before model calls and records prompt-level audits during model execution.

Table 1 reports the main leakage controls. Automatic checks scan visible task fields, document identifiers, titles, source types, timestamps, body text, and visible citations for hidden scorer fields, semantic IDs, material-role labels, and direct-answer cues. They found no critical, high, medium, hidden-label, semantic-ID, or direct-answer-cue hits. A local pre-model surface review covered 400 document rows and 50 unique tasks; it found no critical leaks or direct answer cues. This is not

Table 1: Leakage and prompt-hygiene gate summary for the role-uncued pilot.

Gate	Rows/files	Hits	Status
Automatic leakage audit	480 docs	0 critical/high/medium	Passed
Hidden-label scan	480 docs	0 hidden-label hits	Passed
Semantic-ID/direct-cue scan	480 docs	0 semantic-ID or direct-answer hits	Passed
Local surface review	400 rows	0 critical leaks, 0 direct cues	Passed; local leakage gate only
Prompt audit	480 prompts	0 hidden fields, 0 audit-ID hits	Passed
Stored hidden-label audit	480 outputs	0 prompt hits, 0 output hits	Passed

Table 2: Shortcut baseline summary. Values are operational epistemic escape.

Baseline family	Hidden view	Visible view	Interpretation
ID-only	0.000	0.000	Document IDs carry no useful signal
Cue phrase only	0.000	0.000	Direct cue phrases do not solve the task
Metadata/source/timestamp only	0.000	0.000	Neutral metadata alone is insufficient
Best non-heuristic shortcut	0.133	0.133	Below gate limit
Simple heuristic	0.350	0.333	Strongest shortcut; used for model comparison

independent human-subject validation. It is a pre-model leakage gate used to decide whether model calls are justified.

Table 2 summarizes the no-API shortcuts. The strongest simple heuristic reaches 0.350 operational escape on the hidden view and 0.333 on the visible view. This is high enough to require caution. The correct claim is narrow: direct labels, document IDs, order, metadata alone, and trivial cue phrases do not explain the whole pilot, but row-wise shortcut baselines remain competitive and bound any model comparison. Reproducibility details and baseline definitions are in Appendix B.

5 Metrics And Scoring Contract

The scorer treats final-answer accuracy as one component, not the endpoint. The main row-level outcome is `operational_epistemic_escape`, one scorer contract that asks whether a model both reaches the right epistemic state and preserves a machine-readable evidence/action contract. The full scoring pseudocode is in Appendix A; this section gives the reader-facing contract.

For packet-judgment rows, operational escape requires parse success, correct belief, no polluted document in the support field, and disciplined uncertainty. For evidence-selection rows, it requires parse success, enough selected evidence value, duplicate avoidance, and no generated-lore selection. For active-verification rows, it requires parse success, correct belief, clean support, disciplined uncertainty, and the required action types for the task.

The main component metrics are:

- **Belief correctness**: whether the predicted verdict matches the gold verdict.
- **Evidence precision**: the share of non-empty support that is clean gold support.
- **Clean support recall**: whether available clean support is recovered.
- **Polluted support rate**: how often polluted evidence is used as support.
- **Rejected pollutant rate**: how often known pollutants are explicitly rejected.

Table 3: Main model metrics by neutral metadata view. Operational escape is the joint success metric.

Model	Hidden op.	Hidden belief	Hidden evidence	Visible op.	Visible belief	Visible evidence
gpt-5.5	0.517	1.000	0.758	0.533	0.983	0.750
gemini-3.1-pro-preview	0.417	0.950	0.877	0.450	0.883	0.808
claude-opus-4-7	0.300	0.933	0.500	0.333	0.933	0.463
deepseek-v4-pro	0.217	0.667	0.435	0.250	0.650	0.447

- **Uncertainty discipline**: whether confidence and insufficiency behavior follow the gold state.
- **Verification action score**: whether active-verification actions use the required action type and an executable target.
- **Operational epistemic escape**: the task-family-specific joint success criterion combining parse success, belief, evidence hygiene, uncertainty, and, where needed, action recall.

Operational escape is intentionally stricter than answer accuracy. A model that answers correctly while placing generated lore in `supporting_evidence` should not receive full credit for a structured evidence contract. Likewise, an active-verification answer can be correct while the next action is too vague to execute.

The core-claim audit decomposes operational failures before interpreting them. Each failed row receives one primary component: dirty support, generated-lore selection, evidence value below threshold, duplicate or same-root failure, required action missing, empty support, belief incorrect, parse/schema failure, or other unclassified. This decomposition prevents the old belief-correctness versus operational-escape gap from carrying more meaning than the scorer supports.

We also report a positive evidence-contract check. That check asks whether the output recovers clean support, avoids dirty support, rejects or diagnoses pollutants, and leaves a verification path that can be executed. It is a diagnostic companion to `operational_epistemic_escape`, not a replacement metric.

6 Experimental Setup

The pilot runs four model labels: `gpt-5.5`, `claude-opus-4-7`, `gemini-3.1-pro-preview`, and `deepseek-v4-pro`. Each model receives 120 task instances: 60 latent tasks times two neutral metadata views. The schema variant is `clarified`; the prompt condition is `standard_answer`.

All 480 model records parse successfully. The stored prompt audit reports zero semantic document-ID hits, zero semantic visible-citation hits, zero audit-ID title/body hits, and zero hidden-field hits across 480 prompts. The stored hidden-label audit reports zero prompt hits and zero output hits. The recorded model-call cost is USD 3.764007.

The model-run profiles omit temperature, cap output at 4096 tokens, use first-JSON-object extraction, disable LLM repair, and allow at most two attempts per row. OpenAI, Anthropic, and Google use `json_schema`; DeepSeek uses `json_object` after a provider-specific preflight. Appendix D records the compact invocation table, run date, parser policy, retry policy, and cost cap.

7 Results

Table 3 gives the main model metrics by neutral metadata view. Operational escape is lower than belief correctness for every model and view, but the table should be read descriptively. The pilot is

Table 4: Model metrics with 95% bootstrap intervals from latent-task resampling. Each bootstrap draw resamples the 60 latent tasks and keeps both metadata views paired.

Model	n	Operational escape	Belief correct	Evidence precision
claude-opus-4-7	120	0.317 [0.217, 0.425]	0.933 [0.875, 0.983]	0.481 [0.362, 0.604]
deepseek-v4-pro	120	0.233 [0.142, 0.333]	0.658 [0.558, 0.758]	0.441 [0.323, 0.562]
gemini-3.1-pro-preview	120	0.433 [0.325, 0.550]	0.917 [0.867, 0.958]	0.843 [0.757, 0.918]
gpt-5.5	120	0.525 [0.408, 0.642]	0.992 [0.975, 1.000]	0.754 [0.650, 0.850]

Table 5: Aggregate model metrics by evidence condition across all four models and both views.

Condition	n	Operational	Belief	Evidence precision	Polluted support
Clean	96	0.719	0.927	0.994	0.000
Buried primary	96	0.490	0.979	1.000	0.000
False consensus	96	0.302	0.719	0.659	0.323
Conflicting evidence	96	0.250	0.812	0.479	0.448
Generated lore	96	0.125	0.938	0.000	0.802

too small, and shortcut baselines are too competitive, to support provider ranking from the main text.

Table 4 adds task-level bootstrap intervals. The intervals are wide enough that this pilot should not be read as a fine-grained provider ranking. The visible-minus-hidden paired operational delta is small overall, 0.029 with a 95% bootstrap interval of [-0.021, 0.079], so the two neutral metadata views are best treated as paired diagnostic views rather than separate leaderboards.

Table 5 shows that the aggregate difficulty is not uniform. Clean rows are much easier, with 0.719 operational escape. Generated-lore rows have 0.938 belief correctness but only 0.125 strict operational escape because evidence precision collapses to 0.000 and polluted support remains high under the current support-field contract.

Table 6 reports the same condition-level operational rates with latent-task bootstrap intervals. Generated lore is the clearest stress case for source independence and pollutant allocation: the belief-to-operational diagnostic gap is 0.812 with a 95% interval of [0.714, 0.906].

Table 7 summarizes a diagnostic pattern under the current schema and scorer: correct verdicts can coexist with polluted support, weak source independence, or incomplete evidence-contract behavior. The table is therefore a triage view, not evidence of a schema-independent belief/action separation.

Table 8 decomposes the 433 operational-escape failures across the pilot and Phase 1.3 mini-suite. The two largest primary components are evidence value below threshold and generated-lore selection, each with 126 failures. Belief-incorrect rows account for 72 failures, while dirty support is a smaller but important structured-field failure mode. The decomposition keeps the interpretation local to named scorer components.

Table 9 audits the generated-lore support-field ambiguity behind the strict score. All 72 target generated-lore strict failures were inspected. The audit judged 71 as likely support-field allocation artifacts because the model prose or actions treated the same documents as unverified, derivative, or requiring primary-source tracing rather than clean support. One row remained a likely true dirty-support failure. This supports a sensitivity claim about the current support field, not a claim that all strict failures disappear.

Table 10 gives a positive contract view over 480 pilot rows. Final verdict correctness is much higher than either operational escape or the composite evidence-hygiene contract, showing why the paper reports provenance discipline, pollutant diagnosis, and verification behavior separately.

Table 6: Condition-level operational epistemic escape with 95% bootstrap intervals from latent-task resampling. Values aggregate across models and both neutral metadata views.

Condition	n	Operational escape
`clean`	96	0.719 [0.536, 0.886]
`conflicting_evidence`	96	0.250 [0.083, 0.433]
`false_consensus`	96	0.302 [0.135, 0.486]
`buried_primary`	96	0.490 [0.225, 0.750]
`generated_lore`	96	0.125 [0.056, 0.200]

Table 7: Diagnostic view of final verdict correctness versus operational escape by condition.

Condition	Belief	Operational	Gap	Evidence precision
Generated lore	0.938	0.125	0.812	0.000
Buried primary	0.979	0.490	0.490	1.000
False consensus	0.719	0.302	0.417	0.659
Conflicting evidence	0.812	0.250	0.562	0.479
Clean	0.927	0.719	0.208	0.994

Table 11 bounds the ecological-validity claim. Against row-wise best shortcut baselines, model operational escape is not reliably higher overall: the mean margin is 0.035 with a 95% bootstrap interval of [-0.133, 0.200], and 361 of 480 rows are non-positive. The only condition/family cells clearly above shortcuts in the audit are packet-judgment cells for buried-primary, conflicting-evidence, and false-consensus conditions. Evidence-selection cells are tied or below shortcuts, and generated-lore evidence selection is below shortcuts.

Table 12 summarizes active-verification action behavior. Exact target rate is 1.000 in this pilot, so the remaining variation comes from required action recall, the action score, and whether the active-verification row also satisfies belief and support hygiene. This result is exploratory because the active-verification slice has only 96 rows and shortcut margins are not decisive for active-verification cells.

Table 13 reports a paired Phase 1.3 presentation-perturbation mini-suite over 20 frozen source tasks, two models, and five presentation variants. All 200 calls parsed, and all 40 task-model groups contained the `baseline_original` row plus every perturbation row. In this slice, operational deltas were non-negative for evidence-order randomization, source-type masking, prompt paraphrase, and citation masking. Generated-lore diagnostic gaps were also observed in this slice. Citation masking should be read as a stress test because visible dependency structure can be task-relevant evidence rather than neutral presentation metadata.

8 Schema/Interface Analysis

The current artifact uses the clarified schema for all main results. The earlier cued construction motivated a schema-sensitivity question, but its model results are quarantined and are not used here. After the main role-uncued pilot, we ran a small Phase 1.1 ablation over generated-lore and buried-primary rows. The ablation holds the evidence packet fixed and varies only the structured output interface. It is exploratory follow-up evidence, not a replacement for the Phase 1 main table.

The interface finding that remains valid in this pilot is narrower. Structured fields matter because evidence and action fields can fail while the final verdict succeeds. In generated-lore rows,

Table 8: Primary failure components among operational-escape failures across the pilot and Phase 1.3 mini-suite. Each failed row is assigned one primary component.

Primary failure component	Count	Share of 433 failures
Evidence value below threshold	126	0.291
Generated lore selected	126	0.291
Belief incorrect	72	0.166
Duplicate or same-root failure	64	0.148
Dirty support	33	0.076
Empty support	10	0.023
Required action missing	2	0.005

Table 9: Generated-lore support-field ambiguity sensitivity. The audit inspected all target generated-lore strict failures and treated prose-aware recovery as a sensitivity analysis, not as a replacement scorer.

Audit quantity	Value
Target generated-lore strict failures	72
Prose-aware strict failures becoming passes	71
Gap closed share	0.986
Likely support-field allocation artifacts	71
Likely residual true dirty-support failures	1
Residual row	`uncued_058_visible` / `deepseek-v4-pro`

strict failures are highly sensitive to whether the scorer treats polluted documents in support fields as accepted support or recognizes that prose and actions diagnosed them as unverified. In active-verification rows, the action schema makes executable next steps measurable instead of leaving them as prose.

The core-claim overnight rerun gives a second interface-sensitivity warning. A role-decomposed schema over 144 generated-lore rows changed operational contract success only slightly on average (mean delta 0.014) and did not create a new model ranking. It did, however, make the contract easier to interpret by separating clean support, refuting evidence, rejected or contaminated evidence, diagnostic evidence, and verification actions. This should be read narrowly: schema wording and field semantics are experimental factors for machine-readable evidence roles. They do not solve all strict-score artifacts, explain every Phase 1 gap, or use the quarantined cued results.

9 Human Scorer Audit

Table 14 reports the local scorer-contract audit. The audit reviewed 40 rows balanced across models, views, conditions, task families, and operational successes/failures. It found no scorer disagreements, no systematic scorer bug, and no need for rescoring.

The audit is still limited. It is a local Codex-assisted manual scorer audit, not an independent human-subject review. The claim is therefore restricted to scorer-contract sanity for this pilot, not external validation of every action-quality judgment.

Table 10: Positive evidence-contract summary over 480 pilot rows. The composite contract decomposes evidence hygiene into positive support recovery, pollutant diagnosis, and executable verification behavior.

Metric	Mean
Composite evidence-hygiene contract	0.410
Operational epistemic escape	0.377
Final verdict correct	0.875
No dirty support	0.685
Clean support recovered	0.596
Pollutant rejected or diagnosed	0.817
Verification path executable	1.000

Table 11: Shortcut-boundary audit summary. Margins are model operational escape minus the row-wise best shortcut baseline. Positive cells are interpretable only where the interval is above zero.

Scope	Mean margin	95% CI	Interpretation
Overall	0.035	[-0.133, 0.200]	Not reliably above shortcuts
Non-positive rows	361 / 480	–	Shortcut ties or beats model rows in most cases
Negative rows	102 / 480	–	Shortcut beats the model row
Buried primary / packet judgment	0.675	[0.200, 1.000]	Clearly above shortcuts
Conflicting evidence / packet judgment	0.575	[0.200, 1.000]	Clearly above shortcuts
False consensus / packet judgment	0.625	[0.200, 1.000]	Clearly above shortcuts
Generated lore / evidence selection	-1.000	[-1.000, -1.000]	Below shortcuts

10 Limitations

The pilot is small: 60 latent tasks, two neutral metadata views, four model labels, and one prompt/schema condition. Bootstrap intervals are reported at the latent-task level, but they do not make the pilot statistically powered for fine-grained provider ranking. Reported model differences should be treated as descriptive.

The tasks are synthetic. Synthetic control is useful for isolating source-role failures, but it introduces prose regularities that real evidence environments do not share. The leakage, baseline, and template-diversity audits reduce this risk; they do not eliminate it.

The neutral metadata are synthetic as well. In particular, document timestamps are model-visible but non-evidential, and they should not be interpreted as realistic publication dates, observation dates, recency cues, or answer evidence. False-consensus citation cycles are also synthetic echo-chain structure among polluted documents, not independent corroboration or acyclic provenance trees.

Shortcut baselines are a major validity boundary. Against row-wise best shortcuts, model operational escape is not reliably higher overall, and most rows are non-positive. Above-shortcut evidence is localized to packet-judgment cells in buried-primary, conflicting-evidence, and false-consensus conditions. The paper therefore does not claim broad shortcut resistance, generated-lore superiority over shortcuts, or stable model ranking.

The support-ambiguity audit is local and diagnostic. It found that 71 of 72 target generated-lore strict failures close under prose-aware rescoring, with one residual likely true dirty-support failure. This sensitivity analysis should not be treated as a new main scorer, independent human validation, or evidence about open-web generated content.

The human review evidence has two different scopes. The 400-row pre-model surface review

Table 12: Active-verification action exactness by model. Each model contributes 24 active-verification rows. All columns are computed on the active-verification subset.

Model	Action score	Exact target	Required recall	Active op. escape
claude-opus-4-7	0.875	1.000	0.875	0.292
gemini-3.1-pro-preview	0.812	1.000	0.812	0.417
gpt-5.5	0.792	1.000	0.792	0.542
deepseek-v4-pro	0.688	1.000	0.688	0.167

Table 13: Phase 1.3 paired presentation-perturbation deltas relative to the within-suite baseline. Positive operational deltas mean the perturbation increased operational epistemic escape relative to `baseline_original`; positive polluted-support deltas mean more polluted support.

Variant	Pairs	Δ operational	Δ belief	Δ evidence precision	Δ polluted support
order_randomized	40	0.125	0.075	0.013	-0.075
source_type_masked	40	0.100	0.050	0.026	-0.025
prompt_paraphrase	40	0.100	0.025	0.025	-0.025
citation_masked	40	0.075	-0.075	-0.047	0.075

is a local leakage gate. The 40-row scorer audit is a local Codex-assisted manual audit. Neither should be described as external validation.

The Phase 1.1 schema-ablation slice is small: 16 source tasks, one visible metadata view, two main models, and a local Codex-assisted scorer audit. It supports only an interface-sensitivity claim about machine-readable evidence-role allocation. It should not be treated as a new benchmark result, a provider ranking, or evidence that schema wording explains every Phase 1 failure.

The Phase 1.3 presentation-perturbation mini-suite is also narrow: 20 source tasks, one visible metadata view, two models, and controlled perturbations. It supports only named mini-suite sensitivity claims. In particular, citation masking can remove task-relevant dependency information, so sensitivity under that perturbation should not be interpreted as a benchmark failure by itself. The flip review is local and scorer-assisted rather than an independent human audit.

Finally, the artifact includes hidden labels and gold data for reproducibility, but those files are not model-visible. Any downstream reuse must preserve that separation.

11 Ethics And Scope

Epistemic Hygiene Arena-Uncued is a diagnostic benchmark for controlled polluted evidence environments. It is not a safety certification, a deployment approval test, or a general leaderboard. High performance on this pilot would not establish that an agent is reliable in real legal, medical, scientific, or financial workflows.

The benchmark uses synthetic entities and synthetic evidence packets. This avoids collecting sensitive real-user data and makes gold provenance explicit. The tradeoff is ecological validity: synthetic documents cannot cover the full distribution of open-web evidence, institutional records, or adversarial manipulation.

The intended use is research analysis of failure modes: belief formation, evidence-role hygiene, shortcut sensitivity, and verification-action expression. The intended non-use is automated certification of deployed systems without additional task coverage, independent audits, domain-specific validation, and human oversight.

Table 14: Local scorer-contract audit summary.

Audit item	Value
Reviewed rows	40
Models covered	4, 10 rows each
Views covered	20 hidden, 20 visible
Conditions covered	8 rows each
Families covered	16 packet, 16 evidence, 8 active
Operational successes/failures	20 / 20
Scorer disagreements	0
Systematic scorer bug found	false
Rescoring required	false
Independent human review	false

12 Artifact

The release-facing artifact is rooted at `artifact_uncued_phase1/`. It includes the role-uncued pilot data, prompt artifacts, schemas, scorer files, model outputs, baselines, audits, examples, a minimal reproduction script, and an artifact verifier script. The manifest records 1022 hashed files and explicitly marks `cued_artifacts_included=false`.

The main verifier is:

```
uvruneha-verify-uncued-phase1
--artifact-dir../artifact_uncued_phase1
--reports-dir../reports
--out-dir../reports
```

The artifact verifier passes all package gates: layout, manifest hashes, visible-file hidden-label scan, gold/scorer inclusion, prompt inclusion, leakage gate, baseline gate, local leakage review gate, model-run completeness, scored-result completeness, scorer audit, and explicit status that the main artifact path defers schema ablation. The minimal no-dependency smoke check is `artifact_uncued_phase1/reproduce_minimal.sh`. The schema-ablation follow-up outputs are reported separately under `reports/` and `eha-mvp/results/`; they are not folded into the main pilot artifact.

The older `artifact/` package is quarantined as cued/internal evidence and is not a source for the main results in this paper.

13 Conclusion

Epistemic Hygiene Arena-Uncued shows that, in a controlled role-uncued pilot, final-answer accuracy can hide evidence-contract failures in polluted evidence environments. The central contribution is not a schema-independent belief/action separation. It is a diagnostic framework for asking whether a model preserves provenance discipline, source independence, pollutant rejection, and executable verification behavior when the evidence packet contains generated lore, echo chains, or buried primary records.

The release should be read as a validity-first pilot. It fixes the direct cueing failure exposed by the older run, packages the uncued artifacts, and records leakage, shortcut, model-run, scorer-contract, support-ambiguity, and schema/interface diagnostics. The next step is a larger role-uncued synthetic task set with independent human review and schema/interface choices treated as experimental factors rather than neutral logging details.

A Operational Escape Scoring Specification

For a scored row r, let P_r denote parse success, B_r belief correctness, S_r clean support, and U_r uncertainty discipline. $B_r = 1$ when the predicted verdict equals the gold verdict. $S_r = 1$ when no document in `supporting_evidence` is in the task pollutant set, including contaminant, generated, or stale documents. $U_r = 1$ when the prediction is `insufficient` exactly for gold-insufficient rows and is not `insufficient` otherwise.

The diagnostic support quantities are:

$$\text{evidence precision} = \frac{|\text{supporting evidence} \cap \text{clean support}|}{|\text{supporting evidence}|},$$

when support is non-empty, and

$$\text{clean support recall} = \frac{|\text{supporting evidence} \cap \text{clean support}|}{|\text{gold clean support}|},$$

when gold clean support exists. Pollutant handling is reported through polluted support, rejected pollutant, and generated-lore avoidance metrics; it is not an unreported hidden term.

For packet-judgment rows:

$$O_r = P_r \wedge B_r \wedge S_r \wedge U_r.$$

For evidence-selection rows:

$$O_r = P_r \wedge (\text{evidence value score} \geq 0.50) \wedge (\text{duplicate avoidance} = 1) \wedge (\text{generated lore selected} = 0).$$

For active-verification rows:

$$O_r = P_r \wedge B_r \wedge S_r \wedge U_r \wedge A_r,$$

where A_r requires the relevant primary-opening, contradiction/comparison, and generated-lore trace action types for that task. Exact target rate is reported separately by matching action targets or selected documents to the required document target set.

Pseudocode.

```
pollutants = contaminant_doc_ids + generated_doc_ids + stale_doc_ids
support_dirty = any(doc in pollutants for doc in supporting_evidence)
belief_correct = predicted_verdict == gold_verdict
uncertainty_ok = (
    predicted_verdict == "insufficient"
    if gold_verdict == "insufficient"
    else predicted_verdict != "insufficient"
)

if family == "evidence_selection":
    escape = (
        evidence_value_score >= 0.50
        and duplicate_avoidance_rate == 1.0
        and not generated_lore_selected
    )
elif family == "active_verification":
    escape = belief_correct and not support_dirty and uncertainty_ok and action_ok
else:
    escape = belief_correct and not support_dirty and uncertainty_ok

operational_epistemic_escape = parse_success and escape
```

The worked examples in Appendix C show one successful clean row, one generated-lore row with correct belief but failed support hygiene, and one active-verification row with required action recall.

B Leakage And Shortcut Method Details

The automatic leakage audit scans only model-visible surfaces: task identifiers, questions, document identifiers, titles, source types, timestamps, bodies, and visible citations. It fails the gate for hidden scorer fields, direct answer cues, semantic identifiers that expose construction roles, or forbidden material-role phrases. The pilot audit reports zero critical, high, medium, hidden-label, semantic-ID, and direct-answer-cue hits.

The local surface review sampled 400 document rows covering 50 unique latent tasks. It found zero critical leaks, zero severe leakage rows, and zero role-guess hits. This was a pre-model leakage gate, not an external validation study.

The no-API shortcut baselines are: ID-only, document-order-only, title-only, source-type-only, metadata-only, timestamp-only, citation-graph-only, claim-overlap-only, cue-phrase-only, random-valid-schema, always-insufficient, and simple heuristic. The strongest shortcut is the simple heuristic, at 0.350 operational escape in the hidden view and 0.333 in the visible view. In the row-wise best-shortcut audit, model operational escape is not reliably higher overall than shortcuts (mean margin 0.035, 95% bootstrap interval [-0.133, 0.200]). This means the benchmark is not explained by trivial labels alone, but shortcut competitiveness is a validity boundary and model comparisons should be phrased descriptively.

The relevant reproducibility paths are `eha-mvp/eha/uncued_leakage.py`, `eha-mvp/eha/uncued_baselines.py`, `reports/eha_uncued_leakage_pilot.json`, `reports/eha_uncued_baselines_pilot.json`, and the packaged copies under `artifact_uncued_phase1/`.

C Task And Scoring Examples

The examples below expose scorer-only roles for auditability. These roles were not visible to models; model-visible packets contained only the neutral document fields.

C.1 Clean Or Buried-Primary Success

Task `uncued_000_visible`; family `packet_judgment`, condition `clean`, view `neutral_metadata_visible`.

Question Did Novalis Robotics log 184 completed checks for Atlas valve checks in March 2025?

Model `gemini-3.1-pro-preview`. Gold verdict `supported`; predicted verdict `supported`.

Output fields Support `uncued_000_d03`; selected `uncued_000_d03`; actions [{"action": "search_primary", "target": "monthly ledger for Northline service bay", "rationale": "To verify the 184 completed checks count against the primary source ledger."}].

Scorer decision Belief 1.000; evidence precision 1.000; polluted support 0.000; required action recall n/a; operational escape 1.000.

Answer excerpt Yes, based on the provided evidence, the operations timeline note confirms that the monthly ledger for the Northline service bay lists 184 completed checks for Atlas valve checks in March 2025.

uncued_000_d01 (Service Ledger Extract; timeline note; 2025-01-16; role=context, supports=False): A routing note lists Northline service bay as the site for the Atlas valve checks workstream. The note does not provide a closing count. The note uses the same site code, workstream name, and monthly close...

uncued_000_d03 (Operations Timeline Note; field note; 2025-01-13; role=clean_anchor, supports=True): The monthly ledger for Northline service bay lists 184 completed checks for Atlas valve checks in March 2025. The entry is dated after the close of the month and uses the same work order code as the...

uncued_000_d04 (Program Registry Entry; operations memo; 2025-01-15; role=context, supports=False): A shift memo mentions that Atlas valve checks work was concentrated in the final week of March 2025. It tracks handoff timing but not the final tally. The note uses the same site code, workstream name, and...

uncued_000_d02 (Field Summary Memo; registry entry; 2025-01-14; role=context, supports=False): A planning note for Atlas valve checks describes the inspection route at Northline service bay during March 2025. It gives staffing context and does not include a final monthly count. The note uses the same...

C.2 Correct Belief With Failed Evidence Hygiene

Task uncued_048_visible; family packet_judgment, condition generated_lore, view neutral_metadata_visible.

Question Did Novalis Robotics launch the HelioSplice partner repair program in March 2025?

Model claude-opus-4-7. Gold verdict insufficient; predicted verdict insufficient.

Output fields Support uncued_048_d01,uncued_048_d04; selected uncued_048_d04,uncued_048_d01,uncued_048_d03; actions [{"action": "trace_source", "target": "uncued_048_d01", "rationale": "The roundup is the originating claim; uncued_048_d04 merely cites it. Tracing to a primary operational record is needed to verify the launch."}, {"action": "search_primary", "target":....

Scorer decision Belief 1.000; evidence precision 0.000; polluted support 1.000; required action recall n/a; operational escape 0.000.

Answer excerpt Evidence is insufficient to confirm the launch. The only affirmative mentions (uncued_048_d01 roundup and uncued_048_d04 recap citing it) form a single unsourced chain, while the program registry (uncued_048_d02) and field memo...

uncued_048_d04 (Service Ledger Extract; registry entry; 2025-04-14; role=same_root_echo, supports=False, contamination=unsupported): A partner recap repeats the HelioSplice partner repair program launch wording from the roundup. Its visible citation points to the roundup item. The note uses the same site code, workstream name, and monthly...

uncued_048_d01 (Operations Timeline Note; field note; 2025-04-13; role=unsupported_story, supports=False, contamination=unsupported): A community roundup says the HelioSplice partner repair program appeared during March 2025. The item gives a program name and month but no linked operational record. The note uses the same site code,...

uncued_048_d02 (Program Registry Entry; timeline note; 2025-04-16; role=context, supports=False): A monthly index names the Atlas valve checks workstream and site. It lists regular service categories but no program launch entry. The note uses the same site code, workstream name, and monthly close date as...

uncued_048_d03 (Field Summary Memo; operations memo; 2025-04-15; role=context, supports=False): A service planning memo for Northline service bay lists routine Atlas valve checks work in March 2025. It does not mention the HelioSplice partner repair program. The note uses the same site code, workstream...

C.3 Active-Verification Action Case

Task `uncued_010_visible`; family `active_verification`, condition `clean`, view `neutral_metadata_visible`.

Question Did Novalis Robotics log 118 completed refurbishments for Nova pump refurbishments in July 2025?

Model `claude-opus-4-7`. Gold verdict `supported`; predicted verdict `supported`.

Output fields Support `uncued_010_d01`; selected `uncued_010_d01,uncued_010_d02,uncued_010_d03`; actions [{"action": "trace_source", "target": "uncued_010_d01", "rationale": "Verify the monthly ledger entry's provenance since it is the only document stating the 118 count."}, {"action": "search_primary", "target": "Harbor Annex July 2025 Nova pump refurbishment....

Scorer decision Belief 1.000; evidence precision 1.000; polluted support 0.000; required action recall 1.0; operational escape 1.000.

Answer excerpt Yes—uncued_010_d01 (Field Summary Memo) records 118 completed Nova pump refurbishments at Harbor Annex for July 2025, and no document contradicts this figure. Other documents corroborate the workstream and site context but lack counts....

`uncued_010_d04` (Service Ledger Extract; timeline note; 2025-02-16; role=`context`, supports=False): A routing note lists Harbor Annex as the site for the Nova pump refurbishments workstream. The note does not provide a closing count. The note uses the same site code, workstream name, and monthly close date...
`uncued_010_d03` (Operations Timeline Note; operations memo; 2025-02-15; role=`context`, supports=False): A shift memo mentions that Nova pump refurbishments work was concentrated in the final week of July 2025. It tracks handoff timing but not the final tally. The note uses the same site code, workstream name,...
`uncued_010_d02` (Program Registry Entry; registry entry; 2025-02-14; role=`context`, supports=False): A planning note for Nova pump refurbishments describes the inspection route at Harbor Annex during July 2025. It gives staffing context and does not include a final monthly count. The note uses the same site...
`uncued_010_d01` (Field Summary Memo; field note; 2025-02-13; role=`clean_anchor`, supports=True): The monthly ledger for Harbor Annex lists 118 completed refurbishments for Nova pump refurbishments in July 2025. The entry is dated after the close of the month and uses the same work order code as the...

Required action types: open_best_visible_record. Required target documents by view: neutral_metadata_hidden: uncued_010_d03; neutral_metadata_visible: uncued_010_d01.

D Model Invocation Details

The run recorded 480 complete model rows and no parse failures. The cost report records USD 3.764007 spent against a USD 10 soft cap, USD 15 hard cap, and USD 20 abort cap.

Table 15: Pilot model invocation profiles. Run date: 2026-05-22. All rows used the clarified schema, the standard-answer prompt, a maximum of two attempts, no LLM repair, first-JSON-object extraction, and developer/system content merged into the user message.

Model	Provider	Response format	Temperature	Max output	Parser / retry policy
`gpt-5.5`	OpenAI	`json_schema`	omitted	4096	first JSON object; max 2 attempts
`claude-opus-4-7`	Anthropic	`json_schema`	omitted	4096	first JSON object; max 2 attempts
`gemini-3.1-pro-preview`	Google	`json_schema`	omitted	4096	first JSON object; max 2 attempts
`deepseek-v4-pro`	DeepSeek	`json_object`	omitted	4096	first JSON object; max 2 attempts

E Synthetic Template Diversity Audit

The pilot balances its five conditions at 12 latent tasks each and its task families at 24 packet-judgment, 24 evidence-selection, and 12 active-verification tasks. Each neutral metadata view contains 240 documents. Body length ranges from 45 to 67 words, with median 55 and mean 54.07 words.

In the visible-metadata view, source types are balanced: 60 field notes, 60 operations memos, 60 registry entries, and 60 timeline notes. The hidden-metadata view masks source type to `document` for all 240 documents. Each view has four title strings, 36 timestamps, and 60 documents with visible citations.

The largest prose regularity is intentional: all 240 documents per view contain the neutral relation-control sentence about using the same site code, workstream name, and monthly close date. This prevents relation checks from depending on labels, but it also creates repeated 5-grams across the corpus. We therefore treat the synthetic pilot as a controlled diagnostic slice, not as evidence that template shortcuts are exhausted.

F Phase 1.3 Presentation-Perturbation Mini-Suite

The Phase 1.3 presentation-perturbation mini-suite froze 20 role-uncued source tasks from `eha-mvp/data/uncued-pilot-v1`: five each from `generated_lore`, `buried_primary`, `conflicting_evidence`, and `false_consensus`. Within each condition, the frozen slice used two `packet_judgment` tasks, two `evidence_selection` tasks, and one `active_verification` task where available. The suite used only the `neutral_metadata_visible` view and seed 20260525.

The run compared `baseline_original` against four paired perturbations: `order_randomized`, `source_type_masked`, `prompt_paraphrase`, and `citation_masked`. The schema and scorer were held fixed to the Phase 1 interface. Pre-model gates checked that the selection manifest referenced only the role-uncued pilot data, that prompt parity changed only the intended field for each perturbation, that hidden labels and condition or family labels were absent from model-visible prompts, that projected cost stayed below the hard cap, and that every perturbation had a paired baseline row.

The full run completed 200 planned calls: 20 source tasks times two models times five variants. All 200 latest rows parsed successfully. The cost report recorded USD 2.614767 spent against a USD 7 hard cap. The prompt audit and stored hidden-label audit both passed with zero hits. The paired analysis produced 160 baseline-versus-perturbation comparisons and 57 flip-review rows for local inspection.

The run directory contains the mini-suite result report, paired CSV tables, model-call artifacts, prompt audits, and the flip-review sheet. The top-level reports directory contains the Markdown result summary and the machine-readable JSON summary.

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